Four Wheeler

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Four wheelers old second hand vehicle; Four-wheeler old second-hand Vehicle; Serviceable training Vehicle

What it is used for

A four-wheeler training vehicle is used to teach vehicle inspection, basic maintenance, servicing, and automotive system diagnosis on a serviceable older vehicle. Correct workshop preparation, safe lifting and isolation, and manufacturer-guided procedures help learners develop practical skills while reducing injury and vehicle damage.

Main parts

- Engine and fuel system provide propulsion and engine operation.
- Transmission and drivetrain transfer engine power to the wheels.
- Steering system controls vehicle direction.
- Brake system slows and stops the vehicle.
- Suspension and tyres support the vehicle and maintain road contact.
- Battery and electrical system supply and distribute electrical power.
- Lights, instruments, and warning indicators show vehicle status.
- Vehicle lift, jack points, and wheel chocks support safe workshop access where applicable.
- Diagnostic connector provides access to compatible vehicle diagnostic systems.

Before you start

1. Only trained instructors and authorised learners should perform vehicle servicing under supervision.
2. Confirm the vehicle identity, service history, condition, and available manufacturer service information.
3. Park on firm, level ground, apply the parking brake, select the appropriate gear or Park, and chock the wheels as needed.
4. Inspect the vehicle for fluid leaks, damaged tyres, exposed wiring, and other obvious hazards.
5. Remove the ignition key and isolate the battery when the task requires electrical or mechanical isolation.
6. Use a suitable vehicle lift or approved jack points and rated axle stands for underbody work.
7. Wear suitable workshop clothing, safety footwear, and task-specific PPE; ensure the workshop is ventilated.

How to use

1. Conduct a walk-around inspection and record visible body, glass, tyre, and lighting defects.
2. Check fluid levels, battery condition, belts, hoses, and accessible connections according to the service plan.

3. Identify the engine, transmission, brake, steering, suspension, and electrical components before servicing.
4. Connect a compatible diagnostic tool only when required and follow the vehicle manufacturer's instructions.
5. Perform the assigned inspection or maintenance task using the correct tools and service specifications.
6. For underbody work, raise the vehicle only at approved lift points and verify the support is secure before access.
7. Keep removed fasteners and parts organised and prevent dirt from entering open systems.
8. Reinstall all parts, tighten fasteners to specified torque where required, and restore disconnected connections correctly.
9. Check fluid levels, tool clearance, guards, connectors, and the work area before restarting the vehicle.
10. Start the engine only when safe and check for warning lights, leaks, abnormal noise, or other faults.
11. Conduct the required functional checks or supervised road test only after the vehicle is confirmed safe and legally permitted.
12. Record work completed, measurements, faults, replaced parts, and any follow-up maintenance needed.

⚠ Safety

- Never work beneath a vehicle supported only by a jack; use correctly rated stands or an approved lift.
- Keep clear of moving belts, fans, hot exhaust parts, and rotating components when the engine is running.
- Use suitable exhaust extraction and never run the engine in a closed workshop without adequate ventilation.
- Handle fuel, oils, coolant, brake fluid, and batteries according to their safety instructions.
- Isolate the battery and follow manufacturer procedures before work on electrical or restraint systems.
- Do not road-test an unsafe, unregistered, uninsured, or otherwise unauthorised vehicle.
- Use approved lifting points and never exceed the lift, jack, or stand capacity.

Care & storage

- Clean the vehicle and return tools to their designated storage after the task.
- Inspect tyres, brakes, steering, lights, fluid levels, and visible safety-critical parts at the scheduled interval.
- Dispose of used oil, filters, coolant, batteries, and other workshop waste through approved channels.
- Maintain service records and report defects that could affect vehicle safety.
- Arrange qualified repairs when a defect is outside the learner's authorised scope.

Common mistakes to avoid

- Working beneath a vehicle without rated stands or an approved lift.
- Using incorrect service specifications or tightening fasteners without the required torque guidance.
- Restarting the vehicle with tools, loose parts, or disconnected components remaining.
- Ignoring leaks, warning indicators, tyre defects, or brake and steering problems.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.